

# 物理宇宙学基础作业

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1.(a) 该宇宙模型中的临界密度为:

$$\rho_{\text{crit}} = \frac{3H_0^2}{8\pi G} = 9.2045 \times 10^{-27} \text{ kg/m}^3$$

由题意, 重子物质中 75% 为  $^1\text{H}$ , 25% 为  $^4\text{He}$ , 则当前时刻的电子数密度为:

$$\begin{aligned} n_0 &= n_{\text{H}} + 2n_{\text{He}} \\ &= 75\% \times \frac{\Omega_b \rho_{\text{crit}}}{m_{\text{H}}} + 25\% \times \frac{\Omega_b \rho_{\text{crit}}}{m_{\text{He}}} \\ &= 0.1927 \text{ m}^{-3} \end{aligned}$$

1.(b) 只考虑电子对光子的 Thomson 散射, 散射截面为:

$$\sigma = \frac{1}{6\pi\epsilon_0^2} \left( \frac{e^2}{m_e c^2} \right)^2 = 6.6525 \times 10^{-29} \text{ m}^2$$

光深:

$$\begin{aligned} \tau &= n\sigma l \\ &= n_0(1+z)^3 \cdot \sigma \cdot \frac{r}{1+z} \\ &= n_0\sigma r(1+z)^2 \end{aligned}$$

考虑到:

$$r = \frac{c}{H_0} \int_0^z \frac{dz}{\sqrt{E(z)}} = \frac{c}{H_0} \int_0^z \frac{dz}{\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}}$$

则得光深关于红移的关系:

$$\tau(z) = \frac{n_0\sigma c}{H_0} \int_0^z \frac{(1+z)^2 dz}{\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}}$$

由  $\tau_{\text{CMB}} = 0.09$  及  $\Omega_m + \Omega_\Lambda = 1$  得:

$$\begin{aligned} \tau_{\text{CMB}} &= \frac{n_0\sigma c}{H_0} \int_0^{z_{\text{reion}}} \frac{(1+z)^2 dz}{\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}} \\ &= \frac{n_0\sigma c}{H_0} \int_1^{x_{\text{reion}}} \frac{dx}{\sqrt{\Omega_m x + \Omega_\Lambda}}, \quad x = (1+z)^3 \\ &= \frac{2n_0\sigma c}{3H_0\Omega_m} (\sqrt{\Omega_m x_{\text{reion}}^3 + \Omega_\Lambda} - 1) \\ &= \frac{2n_0\sigma c}{3H_0\Omega_m} (\sqrt{\Omega_m(1+z_{\text{reion}})^3 + \Omega_\Lambda} - 1) \\ &= 0.09 \end{aligned}$$

计算得  $z_{\text{reion}} = 11.33$ , 即宇宙再电离发生于约  $z = 11$  处。

1.(c) 由光深的定义，观测者接收到红移为  $z$  处的辐射为：

$$I = I_0 e^{-\tau}$$

则光被星际介质散射的几率为：

$$P = \frac{I - I_0}{I_0} = 1 - e^{-\tau}$$

由  $\tau = n_0 \sigma r (1+z)^2$  得：

$$\frac{dP}{R_0 d\omega} = \frac{dP}{dr} = \frac{dP}{d\tau} \cdot \frac{d\tau}{dr} == n_0 \sigma (1+z)^2 e^{-\tau(z)}$$

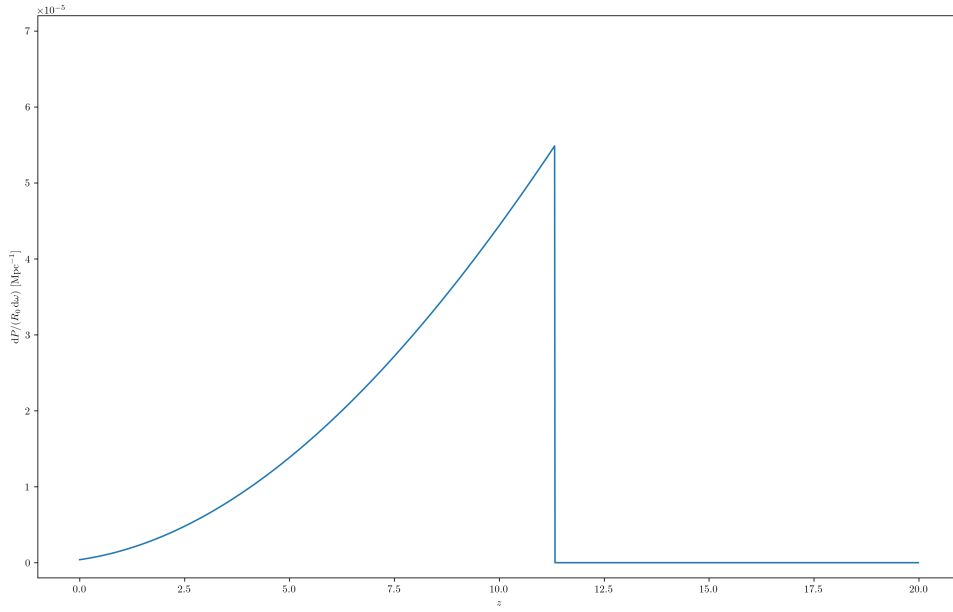


图 1:  $dP/(R_0 d\omega) - z$  关系图

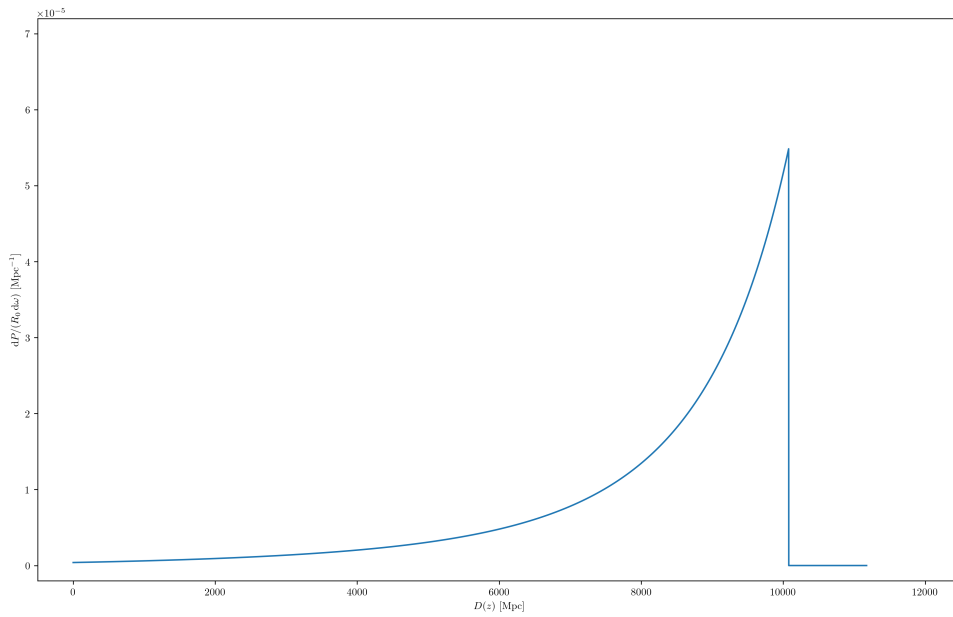


图 2:  $dP/(R_0 d\omega) - D(z)$  关系图，注意此处  $D(z) = r$  即为共动距离